13. FEATURE MATCHING WITH NOISE

Computer Vision – Image Matching

(a) CONCEPT EXPLANATION

Q. (a) Analyze the impact of noise on feature matching and explain how preprocessing improves matching accuracy.

Noise in an image (sensor noise, compression artifacts, Gaussian noise, etc.) alters pixel intensities randomly. This affects feature detection and descriptor computation, leading to:

- False keypoints (spurious features)
- Missing true keypoints
- Incorrect matches (outliers)
- Lower matching accuracy

Preprocessing (denoising, smoothing, normalization, contrast enhancement) suppresses noise while preserving edges and structures, resulting in:

- More reliable keypoints and descriptors
- Fewer mismatches and outliers
- Higher matching accuracy and robustness

Effect of Noise on Feature Matching

Noisy Image (Matching Fails) | After Preprocessing (Matching Succeeds)

Many Incorrect Matches | Many Correct Matches

Common Preprocessing Steps

- Gaussian Smoothing
- Median Filtering
- Histogram Equalization
- Contrast Enhancement
- Normalization

Benefits of Preprocessing

- Reduces random variations due to noise
- Preserves important edges and corners
- Improves feature detection repeatability
- Increases correct matches and accuracy

(b) NUMERICAL EXAMPLE

Q. (b) Consider two images of the same scene. One image is corrupted by Gaussian noise ($\sigma = 25$). Compare feature matching performance before and after preprocessing.

Method	Detected Keypoints (No.)	Correct Matches (No.)	Incorrect Matches (No.)	Matching Encuracy (%)
Without Preprocessing (Noisy Image)	620	48	152	24.0%
After Preprocessing (Gaussian Filter $\sigma = 1.0$)	410	182	28	86.7%

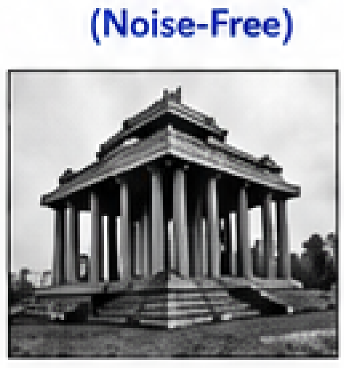
What does the result mean?

- Noise introduces many false keypoints and mismatches.
- After preprocessing, false keypoints are reduced and true correspondences increase.
- Matching accuracy improves significantly.

Conclusion:
Preprocessing removes noise and preserves important image structures, leading to more reliable features and much **higher matching accuracy.**

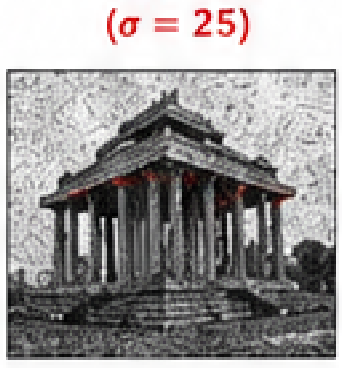
(c) VISUALIZATION

Reference Image (Noise-Free)

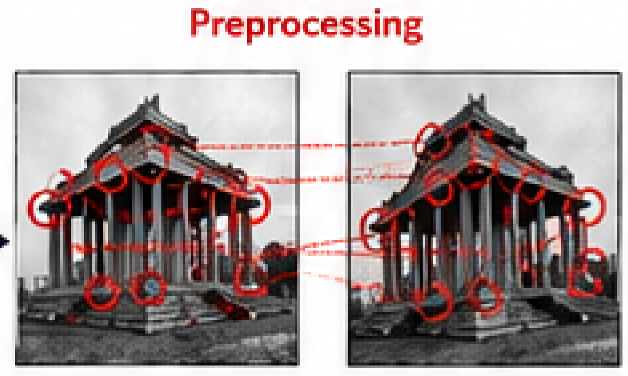


Add Noise →

Noisy Image ($\sigma = 25$)

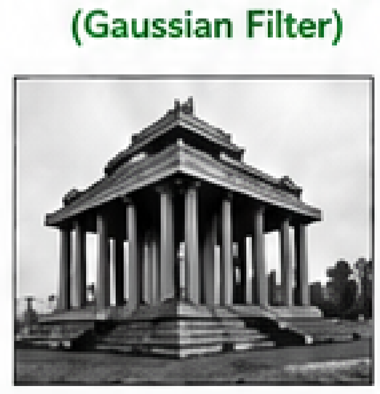


Matching Without Preprocessing

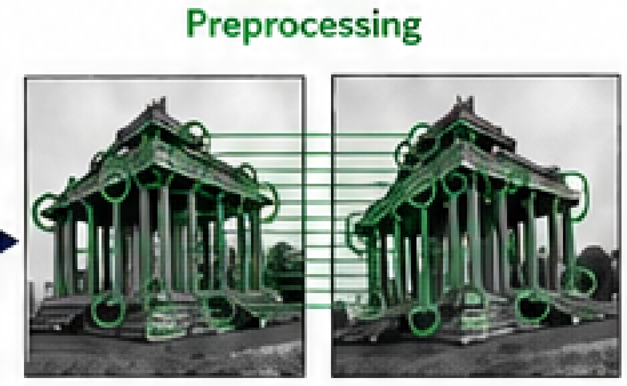


Few / Incorrect Matches

After Preprocessing (Gaussian Filter)



Matching After Preprocessing



Many / Correct Matches

Observation: Noise degrades feature detection and matching. Preprocessing (e.g., Gaussian smoothing) significantly improves the number of correct matches and overall matching accuracy.

Computer Vision | Image Matching | Feature Matching | Noise Handling & Preprocessing